Thm. Let $\{a_n\}$ be a series, and take $R = \left(\limsup_{n\to\infty} |a_n|^{\frac{1}{n}}\right)^{-1}$.

Then, the series $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely if $|x|<R$, diverges if $|x|>R$.

Therefore, we can define $f : (-R, R) \to \mathbb{R} ; x \mapsto \sum_{n=0}^{\infty} a_n x^n$.

For any $0<\varepsilon<R$, f converges uniformly on $[-R+\varepsilon, R-\varepsilon]$ and f is differentiable on $(-R, R)$, with

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

Proof. The first assertion is given by the root test.
And, the second assertion is given by the weierstrass - M test, specifically, for all $n \in \mathbb{N}$ and $x \in [-R+\varepsilon, R-\varepsilon]$, $|a_n x^n| = |a_n|\cdot|x|^n \le |a_n|\cdot|R-\varepsilon|^n$, and

$$\sum_{n=0}^{\infty} |a_n|\cdot|R-\varepsilon|^n$$

converges, by the first. To show the third assertion, observe that

$$\limsup_{n\to\infty} |a_n|^{\frac{1}{n}} = \limsup_{n\to\infty} |n a_n|^{\frac{1}{n}}$$

Because: since $x_n = n^{\frac{1}{n}} - 1$ satisfies $(1+x_n)^n = n \ge \frac{n(n-1)}{2}\cdot x_n^2$, thus

$$x_n^2 \le \frac{2}{n-1} \to 0 \quad \text{as} \quad n\to\infty$$

Thus $n^{\frac{1}{n}} \to 1$ as $n\to\infty$. Given $\varepsilon>0$, there is $N_1 \in \mathbb{N}$ s.t

$$n \ge N_1 \Rightarrow |a_n|^{\frac{1}{n}} < \limsup_{n\to\infty} |a_n|^{\frac{1}{n}} + \frac{\varepsilon}{2}$$

Meanwhile, since $n^{\frac{1}{n}} \to 1$ as $n\to\infty$, take $N_2 \in \mathbb{N}$ s.t

$$n \ge N_2 \Rightarrow n^{\frac{1}{n}} < 1+r, \quad \text{where } r = \frac{A+\varepsilon}{A+\frac{\varepsilon}{2}} - 1, \quad A = \limsup_{n\to\infty} |a_n|^{\frac{1}{n}}.$$

Then,

$$n \ge \max\{N_1, N_2\} \Rightarrow n^{\frac{1}{n}}\cdot |a_n|^{\frac{1}{n}} \le \limsup_{n\to\infty} |a_n|^{\frac{1}{n}} + \varepsilon.$$

Moreover, there are infinitely many $n \in \mathbb{N}$ s.t $\limsup_{n\to\infty} |a_n|^{\frac{1}{n}} - \frac{\varepsilon}{2} < |a_n|^{\frac{1}{n}}$,
we can choose similarly, infinitely many $n \in \mathbb{N}$ s.t $1-t < n^{\frac{1}{n}}$, $t = 1 - \frac{A-\varepsilon}{A-\frac{\varepsilon}{2}}$
Now proved. Using this, $\sum n a_n x^{n-1}$ converges uniformly in $(-R, R)$,
we obtain the result.

「Thm. Suppose that $f:(-R,R)\to\mathbb{R};\ x\mapsto\sum_{n=0}^{\infty}a_n\cdot x^n$ is well-defined.

Given $c\in(-R,R)$, the representation

$$f(x)=\sum_{n=0}^{\infty}\frac{f^{(n)}(c)}{n!}\cdot(x-c)^n \qquad (|x-c|<R-|c|)$$

is valid.

Proof. Given $c\in(-R,R)$,

$$f(x)=\sum_{n=0}^{\infty}a_n\cdot x^n=\sum_{n=0}^{\infty}a_n\cdot[x-c+c]^n=\sum_{n=0}^{\infty}a_n\cdot\sum_{m=0}^{n}\binom{n}{m}\cdot c^{n-m}\cdot(x-c)^m.$$

Meanwhile, since $|x-c|+|c|<R$,

$$\sum_{n=0}^{\infty}\sum_{m=0}^{n}\left|a_n\cdot\binom{n}{m}\cdot c^{n-m}\cdot(x-c)^m\right|=\sum_{n=0}^{\infty}\sum_{m=0}^{n}|a_n|\cdot\binom{n}{m}\cdot|c|^{n-m}\cdot|x-c|^m$$
$$=\sum_{n=0}^{\infty}|a_n|\cdot\big[|x-c|+|c|\big]^n<\infty. \quad \cdots (*)$$

Put
$$A_{nm}=\begin{cases}a_n\cdot\binom{n}{m}\cdot c^{n-m}\cdot(x-c)^m & \text{if } n\geq m\\ 0 & \text{if } n<m.\end{cases}$$
Then,

$\sum_{m=0}^{\infty}|A_{nm}|$ converges for all $n\in\mathbb{N}$, (because for fixed n, $\sum_{m=0}^{\infty}|A_{nm}|$ is just finite sum, actually, $\sum_{m=0}^{\infty}|A_{nm}|=\sum_{m=0}^{n}|A_{nm}|$.) and moreover, $\sum_{n=0}^{\infty}\sum_{m=0}^{\infty}|A_{nm}|$ converges by $(*)$. Now

$$\sum_{n=0}^{\infty}\sum_{m=0}^{\infty}A_{nm}=\sum_{m=0}^{\infty}\sum_{n=0}^{\infty}A_{nm}=\sum_{m=0}^{\infty}\sum_{n=m}^{\infty}A_{nm}=\sum_{m=0}^{\infty}\left[\sum_{n=m}^{\infty}\binom{n}{m}\cdot a_n\cdot c^{n-m}\right]\cdot(x-c)^m$$

Finally, $\frac{1}{m!}\cdot f^{(m)}(x)=\sum_{n=m}^{\infty}\binom{n}{m}\cdot a_n\cdot x^{n-m}$ gives the result, just substituting $x=c$. ☒

※ This Proof is revised at 2026.8.2.

Thm. Suppose that $f : (-R, R) \to \mathbb{R} : x \mapsto \sum_{n=0}^{\infty} a_n \cdot x^n$ is well-defined.

Given $c \in (-R, R)$, the equality holds:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} \cdot (x-c)^n \qquad (x \in B_{R-|c|}(c)).$$

Proof. Observe that

$$f(x) = \sum_{n=0}^{\infty} a_n \cdot [(x-c)+c]^n = \sum_{n=0}^{\infty} a_n \cdot \sum_{m=0}^{n} \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m.$$

And since $x \in B_{R-|c|}(c) \iff |x-c| + |c| < R$,

$$\sum_{n=0}^{\infty} \sum_{m=0}^{n} \left| a_n \cdot \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m \right| = \sum_{n=0}^{\infty} |a_n| \cdot [|x-c| + |c|]^n < \infty.$$

So, $\sum_{m=0}^{\infty} |A_{nm}|$ converges for each n, where $A_{nm} = a_n \cdot \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m$

we can change the limit.

$$f(x) = \sum_{m=0}^{\infty} \left(\sum_{n=m}^{\infty} \binom{n}{m} a_n \cdot c^{n-m} \right) (x-c)^m = \sum_{m=0}^{\infty} \frac{f^{(m)}(c)}{m!} \cdot (x-c)^m.$$

Thm. Given series $\sum a_n$ and $\sum b_n$, define $c_n = \sum_{k=0}^{n} a_k \cdot b_{n-k}$.
If $\sum_{n=0}^{\infty} a_n$ converges absolutely and $\sum_{n=0}^{\infty} b_n$ converges, then

$$\sum_{n=0}^{\infty} c_n = \left(\sum_{n=0}^{\infty} a_n\right) \cdot \left(\sum_{n=0}^{\infty} b_n\right).$$

Proof. Denote

$$A_n = \sum_{k=0}^{n} a_k, \quad B_n = \sum_{k=0}^{n} b_k, \quad B = \sum_{n=0}^{\infty} b_n \quad \text{and} \quad \beta_n = B_n - B$$

Then,

$$\begin{aligned}
\sum_{k=0}^{n} c_k &= a_0 b_0 + (a_0 b_1 + a_1 b_0) + \cdots + (a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0) \\
&= a_0 B_n + a_1 B_{n-1} + \cdots + a_n B_0 \\
&= a_0 (B + \beta_n) + a_1 (B + \beta_{n-1}) + \cdots + a_n (B + \beta_0) \\
&= A_n B + a_0 \beta_n + a_1 \beta_{n-1} + \cdots + a_n \beta_0.
\end{aligned}$$

Now, enough to show that

$$\gamma_n = a_0 \beta_n + a_1 \beta_{n-1} + \cdots + a_n \beta_0$$

Converges to zero.
Let $\varepsilon > 0$ be given. Since $\beta_n \to 0$ as $n \to \infty$, there exists $N \in \mathbb{N}$ s.t $n \geq N \Rightarrow |\beta_n| \leq \varepsilon$.
Now, for $n \geq N$,

$$\begin{aligned}
|\gamma_n| &\leq |\beta_0 a_n + \cdots + \beta_N a_{n-N}| + |\beta_{N+1} a_{n-N+1} + \cdots + \beta_n a_0| \\
&\leq |\beta_0 a_n + \cdots + \beta_N a_{n-N}| + \varepsilon \cdot (|a_{n+N+1}| + \cdots + |a_0|) \\
&\leq |\beta_0 a_n + \cdots + \beta_N a_{n-N}| + \varepsilon \cdot \sum_{k=0}^{\infty} |a_k|
\end{aligned}$$

Using this, Converges absolutely.

Take lim sup, then

$$\begin{aligned}
\limsup_{n\to\infty} |\gamma_n| &\leq \limsup_{n\to\infty} |\beta_0 a_n + \cdots + \beta_N a_{n-N}| + \varepsilon \sum_{k=0}^{\infty} |a_k| \\
&= \varepsilon \cdot \sum_{k=0}^{\infty} |a_k|.
\end{aligned}$$

Consequently, $\gamma_n \to 0$ as $n \to \infty$, so that we obtain the result.

⌈Abel.

⌈Thm. Let $f : (-1,1) \to \mathbb{R} ; x \mapsto \sum_{n=0}^{\infty} a_n x^n$ be defined.

If $\sum_{n=0}^{\infty} a_n$ converges, then $\lim_{x \to 1-} f(x) = \sum_{n=0}^{\infty} a_n$.

Proof. Put $A_n = \sum_{k=0}^{n} a_k$, and $A = \lim_{n \to \infty} A_n = \sum_{n=0}^{\infty} a_n$. As a Cauchy Product,

$$\frac{1}{1-x} f(x) = \sum_{n=0}^{\infty} A_n \cdot x^n, \text{ so } f(x) = (1-x) \cdot \sum_{n=0}^{\infty} A_n x^n \quad (-1<x<1)$$

and since $(1-x) \cdot \sum_{n=0}^{\infty} x_n = 1$, so

$$f(x) - A = (1-x) \sum_{n=0}^{\infty} A_n x^n - A \cdot (1-x) \cdot \sum_{n=0}^{\infty} x_n$$

$$= (1-x)\left(\sum_{n=0}^{\infty} A_n x^n - A \cdot \sum_{n=0}^{\infty} x_n\right)$$

$$= (1-x) \cdot \left(\sum_{n=0}^{\infty} (A_n - A) \cdot x^n\right). \quad (-1<x<1).$$

Given $\varepsilon > 0$, take $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow |A_n - A| < \varepsilon$, and put $M = \max\{|C_n - A| \mid n = 1, \cdots, N\}$.

Now, $|f(x) - A| \leq \left|(1-x) \cdot \sum_{n=0}^{N} (A_n - A) x^n\right| + \left|(1-x) \cdot \sum_{n=N}^{\infty} (A_n - A) x^n\right|$

$$\leq (1-x) \cdot MN + \varepsilon$$

Therefore, $\lim_{x \to 1-} f(x) = A$.

「The Uniqueness Theorem of the Power Series.

「Thm. Suppose that the function of Power series

$$f:(-R,R)\to\mathbb{R}:x\mapsto\sum_{n=0}^{\infty}a_n\cdot x^n$$

is well-defined.

If $Z(f)$ has a limit point in $(-R,R)$, then $a_n=0$ for all $n\geq 0$.

Proof. Let $A=\{x\in(-R,R)\mid x \text{ is the limit point of } Z(f)\}$. ($A$ is non-empty, by the hypo-).

Given $x_0\in A$, the representation

$$f(x)=\sum_{n=0}^{\infty}d_n\cdot(x-x_0)^n \text{ is valid in } |x-x_0|<R-|x_0|.$$

To show that $d_n=0$ for all $n\geq 0$, let k be the smallest non-negative integer s.t. $d_k\neq 0$.

Then, we can write $f(x)=(x-x_0)^k\cdot g(x)$ where $g(x)=\sum_{m=0}^{\infty}d_{k+m}\cdot(x-x_0)^m$. $(|x-x_0|<R-|x_0|)$.

And, the g is continuous at x_0, because the each term is continuous and converges uniformly.

And, $g(x_0)\neq 0$ by the construction. So, for some $\delta>0$, $g(x)\neq 0$ in $|x-x_0|<\delta$.

So, $f(x)\neq 0$ in $0<|x-x_0|<\delta$, it contradicts to x_0 is the limit point of $Z(f)$.

Therefore, $f(x)=0$ for all $|x-x_0|<R-|x_0|$, it means that

$$x_0\in B_{R-|x_0|}(x_0)\subseteq Z(f), \text{ and so } B_{R-|x_0|}\cap(-R,R)\subseteq A, \text{ so } A \text{ is open.}$$

And, the set of limit points is closed, A is the non-empty clopen set of $(-R,R)$.

But, since $(-R,R)$ is connected, the only possibility is $A=(-R,R)$.

Moreover, since f is continuous, $A\subseteq Z(f)$, so $Z(f)=(-R,R)$. ☒